2023-2024
2nd year /Electronic I
Electronic and Telecommunication Department.
Engineering Science/ Kufa University

The Field-Effect Transistor (FET)

MOSFET

3.1 MOS Field-Effect Transistor:

Metal-oxide-semiconductor field-effect transistor (MOSFET)

- The metal-oxide-semiconductor field-effect transistor (MOSFET) becomes a practical reality in the 1970s.
- The MOSFET, compared to BJTs, can be made very small, that is, it occupies a very small area in IC chip.
- In the MOSFET, the current is controlled by an electric field applied perpendicular to both the semiconductor surface and to the direction of current.
- The phenomenon applying an electric field perpendicular to the surface is called the field effect.

3.2 Basic MOS (metal-oxide-semiconductor) structure

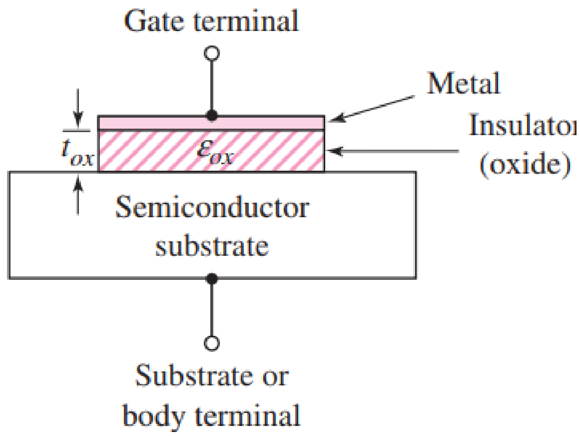

Figure 3.1: Basic MOS structure

3.3 Transistor Structure NMOS

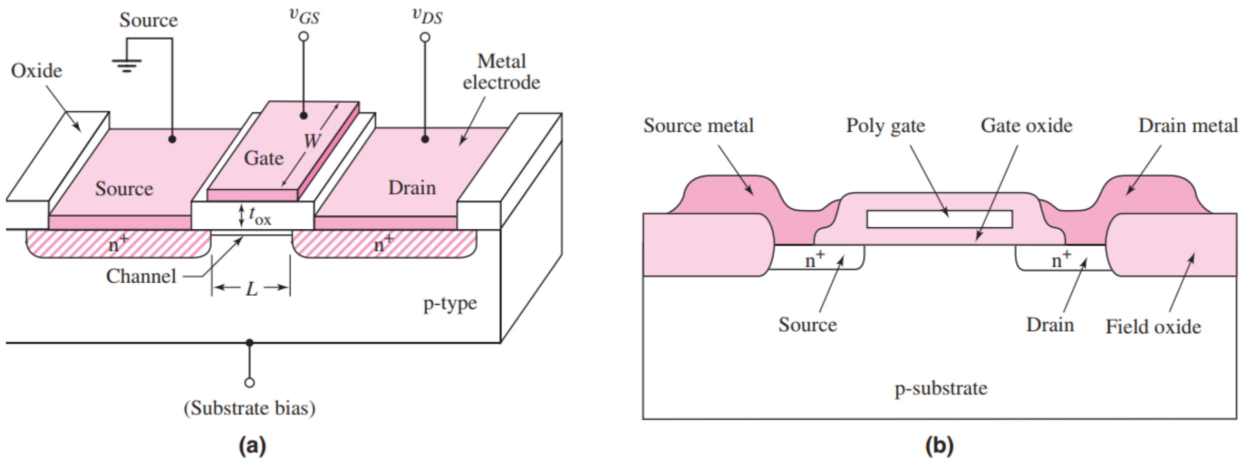

Figure 3.2 (a) schematic diagram of an n-channel -mode MOSFET and (b) an n-channel MOSFET, showing the field oxide and polysilicon gate

- Enhancement mode: a voltage must be applied to the gate to create an inversion layer.
  Electron inversion layer: a positive gate voltage must be applied and for hole inversion layer a negative voltage must be applied.

3.4 Basic transistor operation:

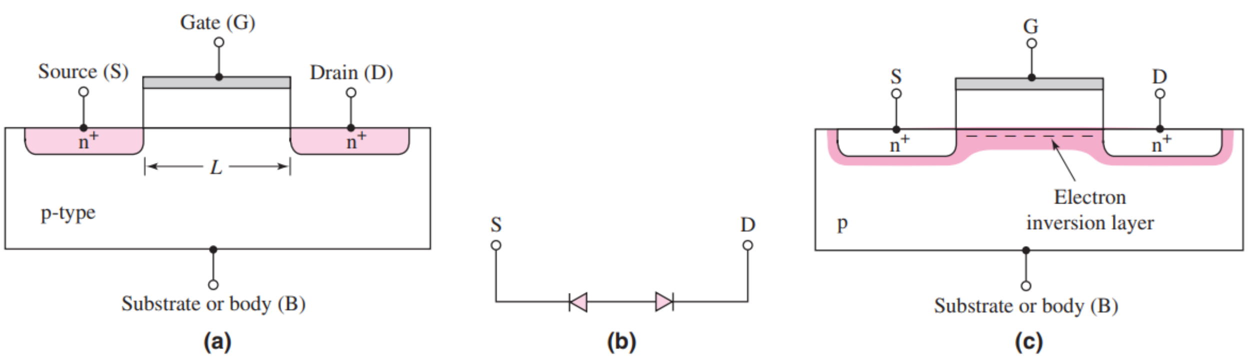

Figure 3.3 (a) Cross section of the n-channel MOSFET (b) equivalent back-to-back diodes between source and drain in cutoff, and (c) inversion layer

The connection between D and S is created so that a current can be generated

3.5 MOSFET current -voltage characteristics:

The threshold voltage of the n-channel MOSFET is denoted as V_{TN} and is defined as the applied gate voltage needed to create an inversion charge.

The threshold voltage as the gate voltage required to turn on the transistor.

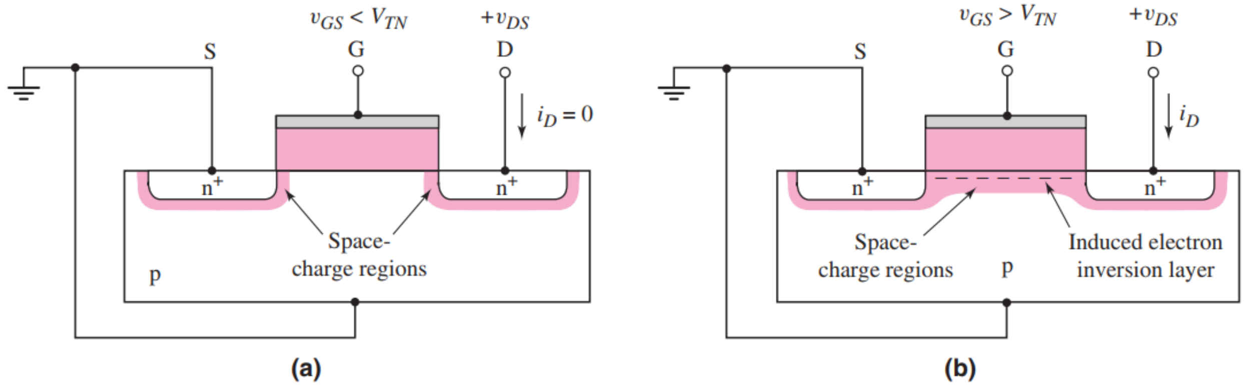

Figure 3.4 The n-channel enhancement -mode MOSFET (a) with an applied gate voltage $V_{GS} < V_{TN}$, and (b) with an applied gate voltage $V_{GS} > V_{TN}$

- The I_D versus V_{DS} characteristics for small values of V_{DS}

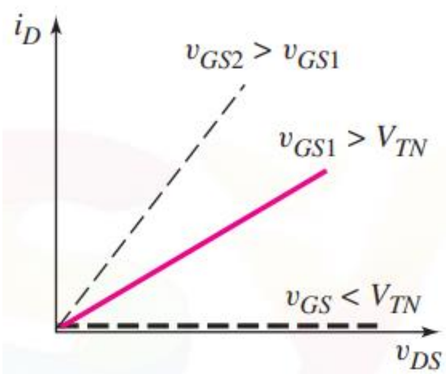

Figure 3.5 Characteristic for I_D versus V_{DS} for small values of V_{DS} at three V_{GS} voltage

- **Nonsaturation (active) or triode region:**

$$v_{DS} < v_{DS(sat)} = v_{GS} - V_{TN}$$

$$i_D = K_n[2(v_{GS} - V_{TN})v_{DS} - v_{DS}^2] = K_n\left(2v_{DS(sat)}v_{DS} - v_{DS}^2\right)$$

- **Saturation region:**

$$v_{DS} > v_{DS(\text{ sat })} (\text{ also } v_{GS} > V_{TN})$$
$$i_D = K_n(v_{GS} - V_{TN})^2$$

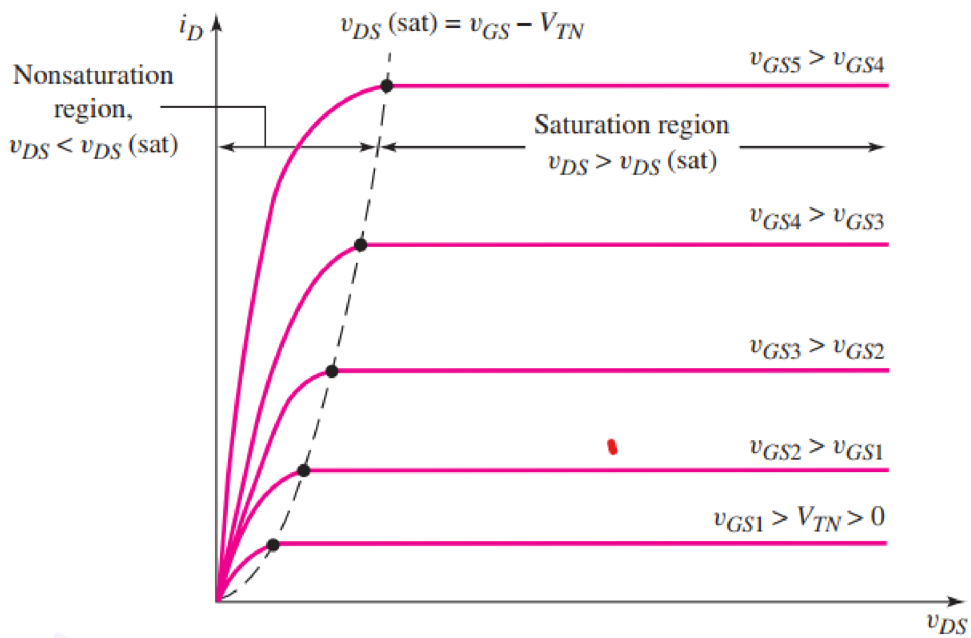

Figure 3.6 Family of i_D versus V_{DS} curves for an n-channel enhancement mode MOSFET

Note: In the saturation region

$$\frac{1}{r_o} = \partial i_D / \partial v_{DS} = \infty$$

- Conduction parameter (K_n):

$$K_n = \frac{W}{L} \cdot \frac{\mu_n C_{ox}}{2}$$

C_{ox} : oxide capacitance per unit area
μ_n : electron mobility
W : channel width
L : channel length

$C_{ox} \propto \frac{1}{t_{ox}}$, t_{ox} : oxide thickness

The conduction parameter is a function of both electrical and geometric parameters.

$$K_n = \frac{W}{L} \cdot \frac{k_n'}{2}$$

k_n': Constant

- **Electrical Parameters**: The oxide capacitance and carrier mobility are constants for a given technology.
- **Geometrical Parameters**: The width-to-length ratio (W/L) is a variable in the design of MOSFETs that is used to produce specific current-voltage characteristics in MOSFET circuits.
- $K_n = \frac{W(\text{cm}) \cdot \mu_n \left(\frac{\text{cm}^2}{\text{V}-\text{s}}\right) \epsilon_{\text{ox}} \left(\frac{\text{F}}{\text{cm}}\right)}{2L(\text{cm}) \cdot t_{\text{ox}}(\text{cm})} = \frac{\text{F}}{\text{V}-\text{s}} = \frac{(\text{C}/\text{V})}{\text{V}-\text{s}} = \frac{\text{A}}{\text{V}^2}$

Ex 3.1:Calculate the current in an n-channel enhancement-mode MOSFET with the following parameters: $V_{TN} = 0.4\text{V}$, $W = 20\ \mu\text{m}$, $L = 0.8\ \mu\text{m}$, $\mu_n = 650\ \text{cm}^2/\text{V–s}$, $t_{ox} = 200$ Å, and $\epsilon_{ox} = (3.9)(8.85 \times 10^{-14})$ F/cm.
Determine the current when the transistor is biased in the saturation region for :
(a) $V_{GS} = 0.8$ V and (b) $V_{GS} = 1.6$ V

Solution: Saturation region

$$i_D = K_n(v_{GS} - V_{TN})^2 \quad \text{and} \quad K_n = \frac{W}{L} \cdot \frac{k_n'}{2} \text{ or } K_n = \frac{W}{L} \cdot \frac{\mu_n C_{ox}}{2}$$

$$K_n = \frac{W \mu_n \epsilon_{\text{ox}}}{2L t_{\text{ox}}}$$

$$K_n = \frac{W \mu_n \epsilon_{\text{ox}}}{2L t_{\text{ox}}} = \frac{(20 \times 10^{-4})(650)(3.9)(8.85 \times 10^{-14})}{2(0.8 \times 10^{-4})(200 \times 10^{-8})}$$

K_n= 1.40 mA/V^2

(a): V_{GS}=0.8V

$$i_D = K_n(v_{GS} - V_{TN})^2 = (1.40)(0.8 - 0.4)^2 = 0.224\text{mA}$$

(b) V_{GS}= 1.6V

$$i_D = (1.40)(1.6 - 0.4)^2 = 2.02\text{mA}$$

3.6 PMOS

- In the p-channel enhancement-mode device, a negative gate-to-source voltage must be applied to create the inversion layer of holes that connects the source and drain regions.

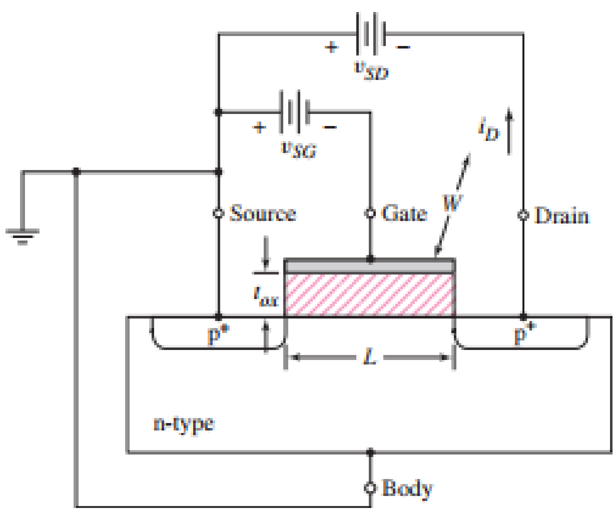

Figure 3.7 Cross section of p-channel enhancement-mode MOSFET. The device is cut off for V_{GS}=0.

- The threshold voltage denoted an V_{TP} for the PMOS is negative for an enhancement mode device. The threshold voltage is positive for a depletion-mode device.
- Holes flow from the source to the drain, the conventional current enters the source and leaves the drain.

- **Nonsaturation (triode) Region**

when : $v_{SD} < v_{SD(sat)} = v_{SG} + V_{TP}$

$$i_D = K_p[2(v_{SG} + V_{TP})v_{SD} - v_{SD}^2] = K_p\left(2v_{SD(sat)}v_{SD} - v_{SD}^2\right)$$

- **Saturation Region**

When
$v_{SD} > v_{SD(sat)}$(also $v_{SG} + V_{TP} > 0$)

$$i_D = K_p(v_{SG} + V_{TP})^2$$

$$i_D = K_p[2(v_{SG} + V_{TP})v_{SD} - v_{SD}^2]$$

$$K_p = \frac{W\mu_p C_{\text{ox}}}{2L}$$

$$K_p = \frac{W}{L}.\frac{k_p'}{2}$$

Ex 3.2 : Determine the source to drain voltage required to bias a p-channel enhancement mode MOFET in the saturation region. Consider K_p= 0.2m A/v^2 , V_{TP} = -0.5 V, and i_D = 0.50 mA.

Solution :

The drain current in the saturating region given as:

$$i_D = K_p(v_{SG} + V_{TP})^2$$

0.50= 02 (v_{SG}-0.5)^2 then V_{SG}= 2.08 V

The voltage is required to bias the p-channel MOSFET in the saturation region:

$$v_{SD} > v_{SD}(\text{sat}) = v_{SG} + V_{TP} = 2.08 - 0.5 = 1.58\text{V}$$

3.7 Circuit Symbols:

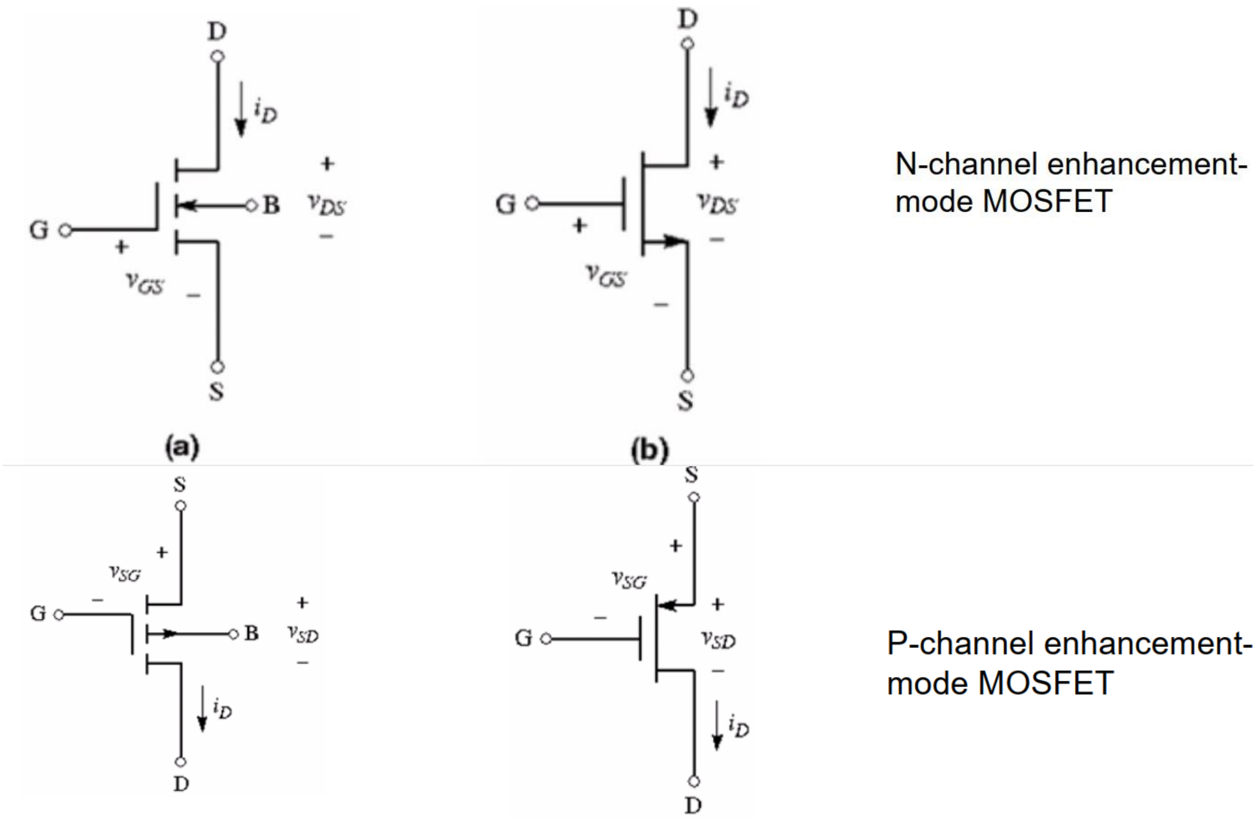

Figure 3.8 Circuit Symbols for (N and P) MOSFET

3.8 Additional MOSFET structures and circuit symbols: Depletion Mode MOSFET

Depletion-Mode MOSFET:

- When zero volts are applied to the gate, an n-channel region (inversion layer) exists under the oxide of impurities introduced during device fabrication.
- The term depletion mode means that a channel exists even at zero gate voltage.
- A negative voltage must be applied to the n-channel depletion-mode MOSFET to turn the device off.

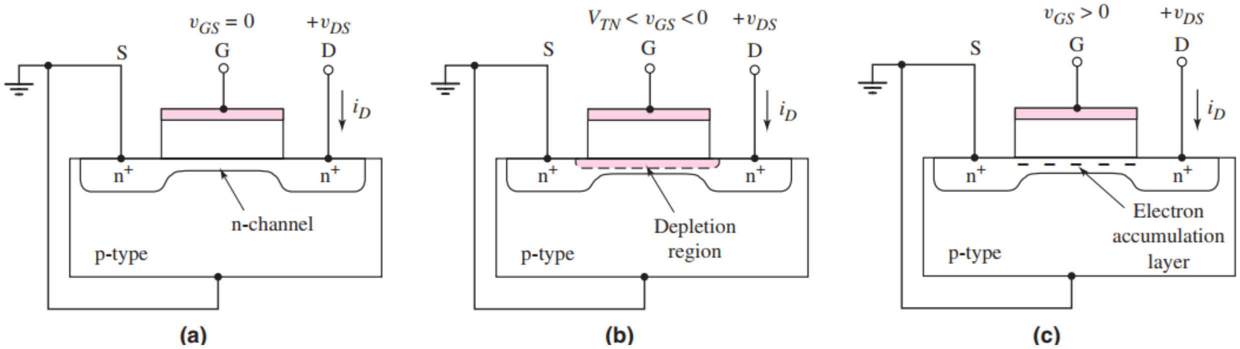

Figure 3.9 Cross section of an n-channel depletion mode MOSFET for: (a) $V_{GS} = 0$,(b) $V_{GS} < 0$, and (c) $V_{GS} > 0$

I-V Curves and Circuit Symbols for Depletion-mode MOSFET

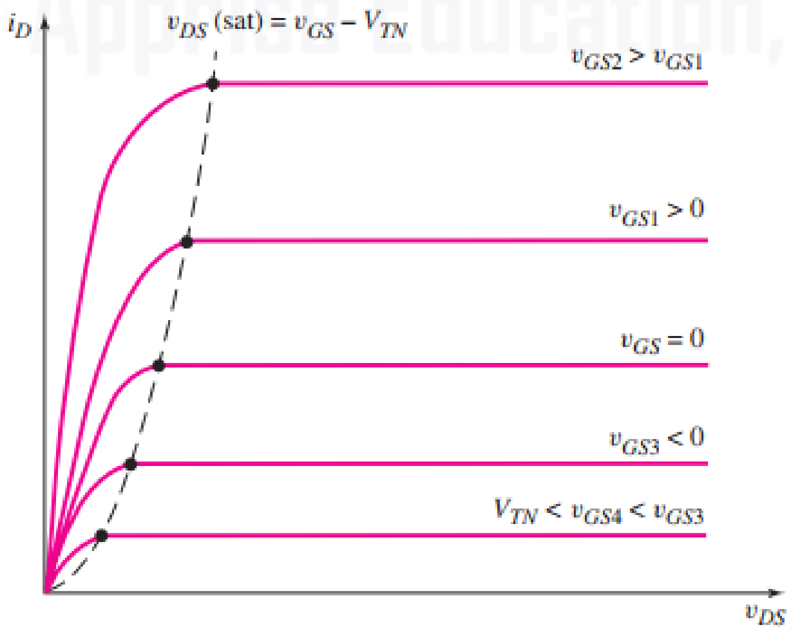

Figure 3.10 The characteristics of n-channel depletion-mode MOSFET

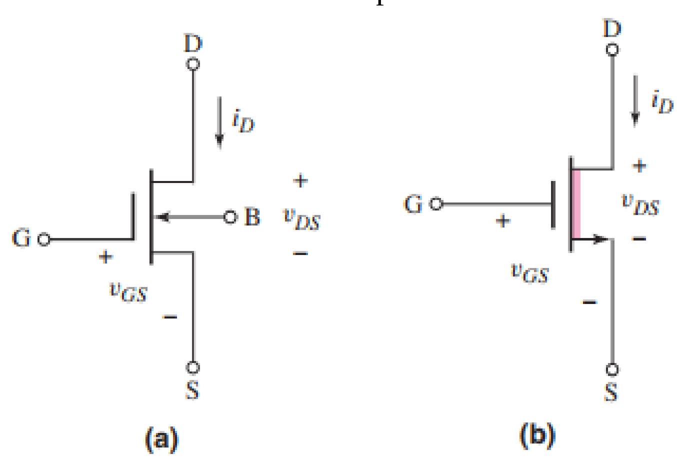

Figure 3.11 The n-channel depletion-mode MOSFET (a) conventional circuit symbol and (b) simplified circuit symbol

3.9 Complementary MOSFETs (CMOS)

- Complement MOS (CMOS) technology uses both NMOS and PMOS in the same circuit.
- To design electrically equivalent NMOS and PMOS devices, adjusting the W/L ratios of the transistors is required.

Figure 3.18 The p-channel depletion mode MOSFET: (a) conventional circuit symbol and (b) simplified circuit symbol

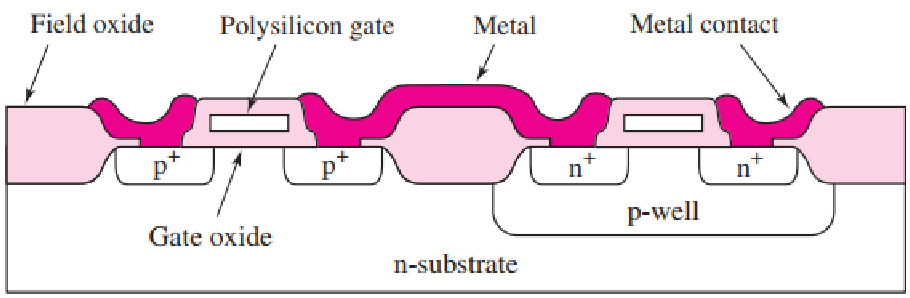

Figure 3.12 Cross section of n-channel transistors fabricated with p-well CMOS technology

Summary of transistor operation

NMOS	PMOS
Nonsaturation region ($v_{DS} < v_{DS}(\text{sat})$)	Nonsaturation region ($v_{SD} < v_{SD}(\text{sat})$)
$i_D = K_n[2(v_{GS} - V_{TN})v_{DS} - v_{DS}^2]$	$i_D = K_p[2(v_{SG} + V_{TP})v_{SD} - v_{SD}^2]$
Saturation region ($v_{DS} > v_{DS}(\text{sat})$)	Saturation region ($v_{SD} > v_{SD}(\text{sat})$)
$i_D = K_n(v_{GS} - V_{TN})^2$	$i_D = K_p(v_{SG} + V_{TP})^2$
Transition point	Transition point
$v_{DS}(\text{sat}) = v_{GS} - V_{TN}$	$v_{SD}(\text{sat}) = v_{SG} + V_{TP}$
Enhancement mode	Enhancement mode
$V_{TN} > 0$	$V_{TP} < 0$
Depletion mode	Depletion mode
$V_{TN} < 0$	$V_{TP} > 0$